Exception Handling and GPIO

1. Reference to the Interrupt Vector Tabel in Appendix A. Declare and code an ISR in C to handle an interrupt request from Port4 pin1. In the ISR, initialize a global variable x of integer data type to 0xFF. Which mode is the processor in when executing the ISR? Assuming that the interrupt was triggered when the processor is executing some user program code, what is the content of the LR register when processor exits the ISR?

-
```
void PORT4_IRQHandler(void); // Declaration

int x;

PORT4_IRQHandler(void) //ISR
{
    x=0xFF;
}
```

- Processor is in Handler mode when processing ISR.
- LR register contains a code word when processor exits the ISR. The value of the code word determines the processor mode and which stack is used after exception return. If the interrupt is triggered when the processor is executing some user program code, then the LR would be either of the two codes that correspond to thread mode execution, since user program is executed in thread mode.
    - 0xFFFFFFF1 Return to handler mode (use main stack)
    - 0xFFFFFFF9 Return to thread mode and use main stack
    - 0xFFFFFFFD Return to thread mode and use process stack

2. The user executes the code in Q1 in Code Composer Studio IDE, without touching the hardware, he halted the execution after 5 seconds. The PC halts somewhere in Default_Handler() routine (see Appendix A). What could be the problem? Or is there a problem?

- Yes, there is a problem with the system or code.
- The interrupt vector table code in Appendix A is setup such that all functions which are not declared explicitly by user is allocated a default label of 'Default_Handler'. Meaning all ISR that are not used will trigger the Default_Handler() function.
- If the PC stops somewhere in the Default_Handler(), that means an interrupt that is not anticipated by the system design has occurred.

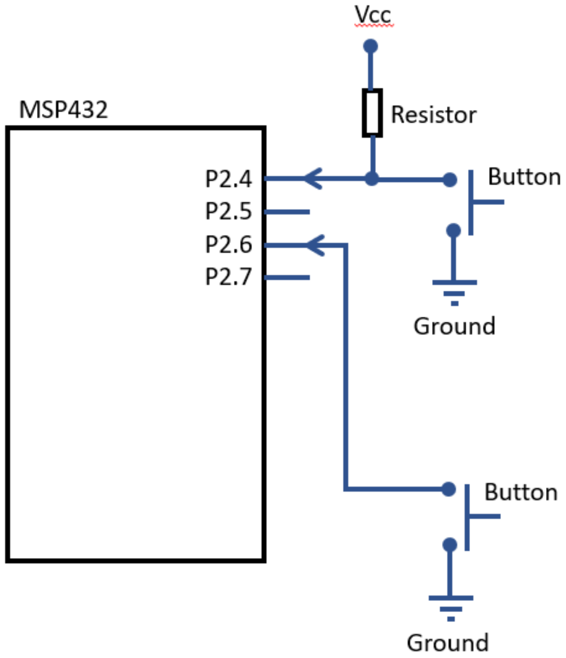
3. The Port 4 Pin1 interrupt request is triggered when the Bump Switch 1 in the system is pressed. When the user perform the same operation as in Q2, the PC stops somewhere in the infinite while loop in the main() – an expected behaviour. But when the user pressed the Bump switch 1 once before halting the processor, the PC is found to stop in the Default_Handler(). What could be the problem? Or is there a problem?

   - Yes, there is a problem with the system or code.
   - The Bump Switch press has triggered an ISR that is not declared in the code, resulting in the direction to Default_Handler() routine.
   - Potential root cause could be a wrong circuit connection between Bump Switch 1 and the processor, or a misaligned interrupt vector table e.g. some vector entries are deleted.
   - Note that all interrupt vector entries need to be in the table, regardless of whether the interrupt is utilized or not.

4. The stacking operation done before entrance to exception handler only save r0-r3 and r12 registers. What should the user software do to handle the context saving/restore of other data registers?

   - Need to look into the AAPCS (ARM Arch Procedure Call Standard) understand how the various data registers are used between main and sub-routines. Exception handler routines can be viewed as a sub-routine.
   - In AAPCS, callee routine i.e. sub-routine is required to save r4-8 and r11, hence the exception service routine user develop should handle the context saving/restore of these registers.
   - Use of r9 is platform specific but in general, there is a requirement the content of r9 has to be persistent across routines. Safest way is to not use r9 in exception service routines.
   - If the exception service routines are coded in C and compiled with a compiler that is AAPCS compliant, the save/restore of the registers will be done automatically by the compiler.

5. The diagram below shows the schematic diagram of a MSP432 connecting to two user buttons on P2.4 and P2.6. The system is designed such that an interrupt will be triggered when the button is asserted.



Perform the necessary registers configuration for the GPIO and Timer peripherals

- GPIO
  - P2.4 and 2.6 port pins initialization
  - Interrupt configuration for Port 2.
  - Set priority of the interrupt to 2.
  - Only need to initialize the following registers
    - For GPIO: PxOUT, PxDIR, PxREN, PxSEL0, PxSEL1, PxIES, PxIE and PxIFG.
    - For interrupt: ISER1, ICPR1, IPR9. Why did we choose these registers for Port2?

The necessary documents have been extracted for you and can be found in the appendix of the tutorial.

- GPIO
  - Port 2.4 and P2.6 (input, internal pullup, falling edge triggered interrupt)
  - GPIO Input
    - P2SEL0 &= ~(0x50), P2SEL1 &= ~(0x50), P2DIR &= ~(0x50).
  - Internal Pullup only needed for P2.6. P2.4 has external pullup.
    - P2REN |= (0x40), P2OUT |= 0x40

    - Interrupt Enable, Clear Interrupt Pending Flag, falling edge triggered.
        - P2IE |= (0x50), P2IFG &= ~(0x50), P2IES |= 0x50

- NVIC
    - Port 2 interrupt. Location [36] in NVIC table, i.e. bit 4 of ISER1.
    - NVIC Interrupt enable
        - ISER1 |= (0x10),
    - Clear any pending bit
        - ICPR1 |= (0x10)
    - Set priority to 2 (only upper 3 bits are valid for MSP432)
        - IPR9 = (IPR9 & 0xFFFFFF00) | 0x00000040

Appendix A

Interrupt Vector Table and related code sample

```
/*******************************************************************************
*
*  MSP432P401R Interrupt Vector Table
*
******************************************************************************/

/* Linker variable that marks the top of the stack. */
extern unsigned long __STACK_END;

/* External declaration for the reset handler that is to be called when the */
/* processor is started                                                     */
extern void _c_int00(void);

/* External declaration for system initialization function                  */
extern void SystemInit(void);

/* Forward declaration of the default fault handlers. */
void Default_Handler            (void) __attribute__((weak));
extern void Reset_Handler       (void) __attribute__((weak));

/* Cortex-M4 Processor Exceptions */
extern void NMI_Handler         (void) __attribute__((weak,
alias("Default_Handler")));
extern void HardFault_Handler   (void) __attribute__((weak,
alias("Default_Handler")));
extern void MemManage_Handler   (void) __attribute__((weak,
alias("Default_Handler")));
extern void BusFault_Handler    (void) __attribute__((weak,
alias("Default_Handler")));
extern void UsageFault_Handler  (void) __attribute__((weak,
alias("Default_Handler")));
extern void SVC_Handler         (void) __attribute__((weak,
alias("Default_Handler")));
extern void DebugMon_Handler    (void) __attribute__((weak,
alias("Default_Handler")));
extern void PendSV_Handler      (void) __attribute__((weak,
alias("Default_Handler")));

/* device specific interrupt handler */
extern void SysTick_Handler     (void)
__attribute__((weak,alias("Default_Handler")));
extern void PSS_IRQHandler      (void)
__attribute__((weak,alias("Default_Handler")));
extern void CS_IRQHandler       (void)
__attribute__((weak,alias("Default_Handler")));
extern void PCM_IRQHandler      (void)
__attribute__((weak,alias("Default_Handler")));
extern void WDT_A_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));
extern void FPU_IRQHandler      (void)
__attribute__((weak,alias("Default_Handler")));
extern void FLCTL_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));
extern void COMP_E0_IRQHandler  (void)
__attribute__((weak,alias("Default_Handler")));
```

```
extern void COMP_E1_IRQHandler  (void)
__attribute__((weak,alias("Default_Handler")));
extern void TA0_0_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));
extern void TA0_N_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));
extern void TA1_0_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));
extern void TA1_N_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));
extern void TA2_0_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));
extern void TA2_N_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));
extern void TA3_0_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));
extern void TA3_N_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));
extern void EUSCIA0_IRQHandler  (void)
__attribute__((weak,alias("Default_Handler")));
extern void EUSCIA1_IRQHandler  (void)
__attribute__((weak,alias("Default_Handler")));
extern void EUSCIA2_IRQHandler  (void)
__attribute__((weak,alias("Default_Handler")));
extern void EUSCIA3_IRQHandler  (void)
__attribute__((weak,alias("Default_Handler")));
extern void EUSCIB0_IRQHandler  (void)
__attribute__((weak,alias("Default_Handler")));
extern void EUSCIB1_IRQHandler  (void)
__attribute__((weak,alias("Default_Handler")));
extern void EUSCIB2_IRQHandler  (void)
__attribute__((weak,alias("Default_Handler")));
extern void EUSCIB3_IRQHandler  (void)
__attribute__((weak,alias("Default_Handler")));
extern void ADC14_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));
extern void T32_INT1_IRQHandler (void)
__attribute__((weak,alias("Default_Handler")));
extern void T32_INT2_IRQHandler (void)
__attribute__((weak,alias("Default_Handler")));
extern void T32_INTC_IRQHandler (void)
__attribute__((weak,alias("Default_Handler")));
extern void AES256_IRQHandler   (void)
__attribute__((weak,alias("Default_Handler")));
extern void RTC_C_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));
extern void DMA_ERR_IRQHandler  (void)
__attribute__((weak,alias("Default_Handler")));
extern void DMA_INT3_IRQHandler (void)
__attribute__((weak,alias("Default_Handler")));
extern void DMA_INT2_IRQHandler (void)
__attribute__((weak,alias("Default_Handler")));
extern void DMA_INT1_IRQHandler (void)
__attribute__((weak,alias("Default_Handler")));
extern void DMA_INT0_IRQHandler (void)
__attribute__((weak,alias("Default_Handler")));
extern void PORT1_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));
```

```
extern void PORT2_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));
extern void PORT3_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));
extern void PORT4_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));
extern void PORT5_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));
extern void PORT6_IRQHandler    (void)
__attribute__((weak,alias("Default_Handler")));

/* Interrupt vector table.  */
#pragma RETAIN(interruptVectors)
#pragma DATA_SECTION(interruptVectors, ".intvecs")
void (* const interruptVectors[])(void) =
{
    (void (*)(void))((uint32_t)&__STACK_END),
                                           /* The initial stack pointer */
    Reset_Handler,                         /* The reset handler         */
    NMI_Handler,                           /* The NMI handler           */
    HardFault_Handler,                     /* The hard fault handler    */
    MemManage_Handler,                     /* The MPU fault handler     */
    BusFault_Handler,                      /* The bus fault handler     */
    UsageFault_Handler,                    /* The usage fault handler   */
    0,                                     /* Reserved                  */
    0,                                     /* Reserved                  */
    0,                                     /* Reserved                  */
    0,                                     /* Reserved                  */
    SVC_Handler,                           /* SVCall handler            */
    DebugMon_Handler,                      /* Debug monitor handler     */
    0,                                     /* Reserved                  */
    PendSV_Handler,                        /* The PendSV handler        */
    SysTick_Handler,                       /* The SysTick handler       */
    PSS_IRQHandler,                        /* PSS ISR                   */
    CS_IRQHandler,                         /* CS ISR                    */
    PCM_IRQHandler,                        /* PCM ISR                   */
    WDT_A_IRQHandler,                      /* WDT ISR                   */
    FPU_IRQHandler,                        /* FPU ISR                   */
    FLCTL_IRQHandler,                      /* FLCTL ISR                 */
    COMP_E0_IRQHandler,                    /* COMP0 ISR                 */
    COMP_E1_IRQHandler,                    /* COMP1 ISR                 */
    TA0_0_IRQHandler,                      /* TA0_0 ISR                 */
    TA0_N_IRQHandler,                      /* TA0_N ISR                 */
    TA1_0_IRQHandler,                      /* TA1_0 ISR                 */
    TA1_N_IRQHandler,                      /* TA1_N ISR                 */
    TA2_0_IRQHandler,                      /* TA2_0 ISR                 */
    TA2_N_IRQHandler,                      /* TA2_N ISR                 */
    TA3_0_IRQHandler,                      /* TA3_0 ISR                 */
    TA3_N_IRQHandler,                      /* TA3_N ISR                 */
    EUSCIA0_IRQHandler,                    /* EUSCIA0 ISR               */
    EUSCIA1_IRQHandler,                    /* EUSCIA1 ISR               */
    EUSCIA2_IRQHandler,                    /* EUSCIA2 ISR               */
    EUSCIA3_IRQHandler,                    /* EUSCIA3 ISR               */
    EUSCIB0_IRQHandler,                    /* EUSCIB0 ISR               */
    EUSCIB1_IRQHandler,                    /* EUSCIB1 ISR               */
    EUSCIB2_IRQHandler,                    /* EUSCIB2 ISR               */
    EUSCIB3_IRQHandler,                    /* EUSCIB3 ISR               */
    ADC14_IRQHandler,                      /* ADC14 ISR                 */
    T32_INT1_IRQHandler,                   /* T32_INT1 ISR              */
```

```
    T32_INT2_IRQHandler,                       /* T32_INT2 ISR                */
    T32_INTC_IRQHandler,                       /* T32_INTC ISR                */
    AES256_IRQHandler,                         /* AES ISR                     */
    RTC_C_IRQHandler,                          /* RTC ISR                     */
    DMA_ERR_IRQHandler,                        /* DMA_ERR ISR                 */
    DMA_INT3_IRQHandler,                       /* DMA_INT3 ISR                */
    DMA_INT2_IRQHandler,                       /* DMA_INT2 ISR                */
    DMA_INT1_IRQHandler,                       /* DMA_INT1 ISR                */
    DMA_INT0_IRQHandler,                       /* DMA_INT0 ISR                */
    PORT1_IRQHandler,                          /* PORT1 ISR                   */
    PORT2_IRQHandler,                          /* PORT2 ISR                   */
    PORT3_IRQHandler,                          /* PORT3 ISR                   */
    PORT4_IRQHandler,                          /* PORT4 ISR                   */
    PORT5_IRQHandler,                          /* PORT5 ISR                   */
    PORT6_IRQHandler                           /* PORT6 ISR                   */
};

                                                              */
void Reset_Handler(void)
{
    SystemInit();

    /* Jump to the CCS C Initialization Routine. */
    __asm("    .global _c_int00\n"
          "    b.w     _c_int00");
}
                                                  */
void Default_Handler(void)
{
      /* Enter an infinite loop. */
      while(1)
      {
      }
}
```

Appendix B

GPIO

Table 6-67. Port P2 (P2.4 to P2.7) Pin Functions

PIN NAME (P2.x)	x	FUNCTION	CONTROL BITS OR SIGNALS[1]			
			P2DIR.x	P2SEL1.x	P2SEL0.x	P2MAPx
P2.4/PM_TA0.1[2]	4	P2.4 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI1A	0	0	1	default
		TA0.1	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0	1	1	X
		DVSS	1			
P2.5/PM_TA0.2[2]	5	P2.5 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI2A	0	0	1	default
		TA0.2	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0	1	1	X
		DVSS	1			
P2.6/PM_TA0.3[2]	6	P2.6 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI3A	0	0	1	default
		TA0.3	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0	1	1	X
		DVSS	1			
P2.7/PM_TA0.4[2]	7	P2.7 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI4A	0	0	1	default
		TA0.4	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0	1	1	X
		DVSS	1			

(1) X = don't care
(2) Not available on the 64-pin RGC package.

12.4.3 PxOUT Register

Port X Output Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-3. PxOUT Register

7	6	5	4	3	2	1	0
PxOUT							
rw	rw	rw	rw	rw	rw	rw	rw

Table 12-6. PxOUT Register Description

Bit	Field	Type	Reset	Description
7-0	PxOUT	RW	Undefined	Port X output. When I/O configured to output mode: 0b = Output is low. 1b = Output is high. When I/O configured to input mode and pullups/pulldowns enabled: 0b = Pulldown selected 1b = Pullup selected

12.4.4 *PxDIR Register*

Port X Direction Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-4. PxDIR Register

7	6	5	4	3	2	1	0
PxDIR							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-7. PxDIR Register Description

Bit	Field	Type	Reset	Description
7-0	PxDIR	RW	0h	Port X direction. 0b = Port configured as input 1b = Port configured as output

12.4.5 *PxREN Register*

Port X Pullup or Pulldown Resistor Enable Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-5. PxREN Register

7	6	5	4	3	2	1	0
PxREN							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-8. PxREN Register Description

Bit	Field	Type	Reset	Description
7-0	PxREN	RW	0h	Port X pullup or pulldown resistor enable. When the port is configured as an input, setting this bit enables or disables the pullup or pulldown. 0b = Pullup or pulldown disabled 1b = Pullup or pulldown enabled

12.4.7 *PxSEL0 Register*

Port X Function Selection Register 0 (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-7. PxSEL0 Register

7	6	5	4	3	2	1	0
PxSEL0							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-10. PxSEL0 Register Description

Bit	Field	Type	Reset	Description
7-0	PxSEL0	RW	0h	Port function selection. Each bit corresponds to one channel on Port X. The values of each bit position in PxSEL1 and PxSEL0 are combined to specify the function. For example, if P1SEL1.5 = 1 and P1SEL0.5 = 0, then the secondary module function is selected for P1.5. See PxSEL1 for the definition of each value.

12.4.8 PxSEL1 Register

Port X Function Selection Register 1 (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-8. PxSEL1 Register

7	6	5	4	3	2	1	0
PxSEL1							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-11. PxSEL1 Register Description

Bit	Field	Type	Reset	Description
7-0	PxSEL1	RW	0h	Port function selection. Each bit corresponds to one channel on Port X. The values of each bit position in PxSEL1 and PxSEL0 are combined to specify the function. For example, if P1SEL1.5 = 1 and P1SEL0.5 = 0, then the secondary module function is selected for P1.5. 00b = General-purpose I/O is selected 01b = Primary module function is selected 10b = Secondary module function is selected 11b = Tertiary module function is selected

12.4.10 PxIES Register

Port X Interrupt Edge Select Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10)

Figure 12-10. PxIES Register

7	6	5	4	3	2	1	0
PxIES							
rw	rw	rw	rw	rw	rw	rw	rw

Table 12-13. P1IES Register Description

Bit	Field	Type	Reset	Description
7-0	PxIES	RW	Undefined	Port X interrupt edge select 0b = PxIFG flag is set with a low-to-high transition. 1b = PxIFG flag is set with a high-to-low transition.

12.4.11 PxIE Register

Port X Interrupt Enable Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10)

Figure 12-11. PxIE Register

7	6	5	4	3	2	1	0
PxIE							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-14. PxIE Register Description

Bit	Field	Type	Reset	Description
7-0	PxIE	RW	0h	Port X interrupt enable 0b = Corresponding port interrupt disabled 1b = Corresponding port interrupt enabled

12.4.12 PxIFG Register

Port X Interrupt Flag Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10)

Figure 12-12. PxIFG Register

7	6	5	4	3	2	1	0
PxIFG							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-15. PxIFG Register Description

Bit	Field	Type	Reset	Description
7-0	PxIFG	RW	0h	Port X interrupt flag 0b = No interrupt is pending. 1b = Interrupt is pending.

NVIC

NVIC Table (P118 MSP432 Datasheets)

Table 6-39. NVIC Interrupts (continued)

NVIC INTERRUPT INPUT	SOURCE	FLAGS IN SOURCE
INTISR[4]	FPU_INT [2]	Combined interrupt from flags in the FPSCR (part of Cortex-M4 FPU)
INTISR[5]	FLCTL	Flash Controller interrupt flags
INTISR[6]	COMP_E0	Comparator_E0 interrupt flags
INTISR[7]	COMP_E1	Comparator_E1 interrupt flags
INTISR[8]	Timer_A0	TA0CCTL0.CCIFG
INTISR[9]	Timer_A0	TA0CCTLx.CCIFG (x = 1 to 4), TA0CTL.TAIFG
INTISR[10]	Timer_A1	TA1CCTL0.CCIFG
INTISR[11]	Timer_A1	TA1CCTLx.CCIFG (x = 1 to 4), TA1CTL.TAIFG
INTISR[12]	Timer_A2	TA2CCTL0.CCIFG
INTISR[13]	Timer_A2	TA2CCTLx.CCIFG (x = 1 to 4), TA2CTL.TAIFG
INTISR[14]	Timer_A3	TA3CCTL0.CCIFG
INTISR[15]	Timer_A3	TA3CCTLx.CCIFG (x = 1 to 4), TA3CTL.TAIFG
INTISR[16]	eUSCI_A0	UART or SPI mode TX, RX, and Status Flags
INTISR[17]	eUSCI_A1	UART or SPI mode TX, RX, and Status Flags
INTISR[18]	eUSCI_A2	UART or SPI mode TX, RX, and Status Flags
INTISR[19]	eUSCI_A3	UART or SPI mode TX, RX, and Status Flags
INTISR[20]	eUSCI_B0	SPI or I^2C mode TX, RX, and Status Flags (I^2C in multiple-slave mode)
INTISR[21]	eUSCI_B1	SPI or I^2C mode TX, RX, and Status Flags (I^2C in multiple-slave mode)
INTISR[22]	eUSCI_B2	SPI or I^2C mode TX, RX, and Status Flags (I^2C in multiple-slave mode)
INTISR[23]	eUSCI_B3	SPI or I^2C mode TX, RX, and Status Flags (I^2C in multiple-slave mode)
INTISR[24]	Precision ADC	IFG[0-31], LO/IN/HI-IFG, RDYIFG, OVIFG, TOVIFG
INTISR[25]	Timer32_INT1	Timer32 interrupt for Timer1
INTISR[26]	Timer32_INT2	Timer32 interrupt for Timer2
INTISR[27]	Timer32_INTC	Timer32 Combined Interrupt
INTISR[28]	AES256	AESRDYIFG
INTISR[29]	RTC_C	OFIFG, RDYIFG, TEVIFG, AIFG, RT0PSIFG, RT1PSIFG
INTISR[30]	DMA_ERR	DMA error interrupt
INTISR[31]	DMA_INT3	DMA completion interrupt3
INTISR[32]	DMA_INT2	DMA completion interrupt2
INTISR[33]	DMA_INT1	DMA completion interrupt1
INTISR[34]	DMA_INT0 [3]	DMA completion interrupt0
INTISR[35]	I/O Port P1	P1IFG.x (x = 0 to 7)
INTISR[36]	I/O Port P2	P2IFG.x (x = 0 to 7)
INTISR[37]	I/O Port P3	P3IFG.x (x = 0 to 7)
INTISR[38]	I/O Port P4	P4IFG.x (x = 0 to 7)
INTISR[39]	I/O Port P5	P5IFG.x (x = 0 to 7)
INTISR[40]	I/O Port P6	P6IFG.x (x = 0 to 7)
INTISR[41]	Reserved	
INTISR[42]	Reserved	

2.4.3.2 ISER1 Register (Offset = 104h) [reset = 00000000h]

ISER1 is shown in Figure 2-21 and described in Table 2-27.

Irq 32 to 63 Set Enable Register. Use the Interrupt Set-Enable Registers to enable interrupts and determine which interrupts are currently enabled.

Figure 2-21. ISER1 Register

<table>
<tr><td>31</td><td>30</td><td>29</td><td>28</td><td>27</td><td>26</td><td>25</td><td>24</td><td>23</td><td>22</td><td>21</td><td>20</td><td>19</td><td>18</td><td>17</td><td>16</td><td>15</td><td>14</td><td>13</td><td>12</td><td>11</td><td>10</td><td>9</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
<tr><td colspan="32">SETENA</td></tr>
<tr><td colspan="32">R/W-0h</td></tr>
</table>

Table 2-27. ISER1 Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	SETENA	R/W	0h	Writing 0 to a SETENA bit has no effect, writing 1 to a bit enables the corresponding interrupt. Reading the bit returns its current enable state. Reset clears the SETENA fields.

2.4.3.8 ICPR1 Register (Offset = 284h) [reset = 00000000h]

ICPR1 is shown in Figure 2-27 and described in Table 2-33.

Irq 32 to 63 Clear Pending Register. Use the Interrupt Clear-Pending Registers to clear pending interrupts and determine which interrupts are currently pending.

Figure 2-27. ICPR1 Register

<table>
<tr><td>31</td><td>30</td><td>29</td><td>28</td><td>27</td><td>26</td><td>25</td><td>24</td><td>23</td><td>22</td><td>21</td><td>20</td><td>19</td><td>18</td><td>17</td><td>16</td><td>15</td><td>14</td><td>13</td><td>12</td><td>11</td><td>10</td><td>9</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
<tr><td colspan="32">CLRPEND</td></tr>
<tr><td colspan="32">R/W-0h</td></tr>
</table>

Table 2-33. ICPR1 Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	CLRPEND	R/W	0h	Writing 0 to a CLRPEND bit has no effect, writing 1 to a bit clears the corresponding pending interrupt. Reading the bit returns its current state.

2.4.3.20 IPR9 Register (Offset = 424h) [reset = 00000000h]

IPR9 is shown in Figure 2-39 and described in Table 2-45.

Irq 36 to 39 Priority Register. Use the Interrupt Priority Registers to assign a priority from 0 to 255 to each of the available interrupts. 0 is the highest priority, and 255 is the lowest.

The hardware priority mechanism only looks at the upper N bits of the priority field (where N is 3 for the MSP432 family), so any prioritization must be performed in those bits. The remaining bits can be used to sub-prioritize the interrupt sources, and may be used by the hardware priority mechanism on a future part

Figure 2-39. IPR9 Register

<table>
<tr><td>31</td><td>30</td><td>29</td><td>28</td><td>27</td><td>26</td><td>25</td><td>24</td><td>23</td><td>22</td><td>21</td><td>20</td><td>19</td><td>18</td><td>17</td><td>16</td><td>15</td><td>14</td><td>13</td><td>12</td><td>11</td><td>10</td><td>9</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
<tr><td colspan="8">PRI_39</td><td colspan="8">PRI_38</td><td colspan="8">PRI_37</td><td colspan="8">PRI_36</td></tr>
<tr><td colspan="8">R/W-0h</td><td colspan="8">R/W-0h</td><td colspan="8">R/W-0h</td><td colspan="8">R/W-0h</td></tr>
</table>

Table 2-45. IPR9 Register Field Descriptions

Bit	Field	Type	Reset	Description
31-24	PRI_39	R/W	0h	Priority of interrupt 39
23-16	PRI_38	R/W	0h	Priority of interrupt 38
15-8	PRI_37	R/W	0h	Priority of interrupt 37
7-0	PRI_36	R/W	0h	Priority of interrupt 36